Maxwell T. Elling

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Personal Site · Google Scholar · ResearchGate

Education

PhD in Atmospheric and Oceanic Sciences 2024–2027 (expected)
University of Colorado Boulder; Advisor: Kris Karnauskas

MA in Climate and Society 2020–2021
Columbia University

BA in Financial Economics 2015–2019
Columbia University

Research Experience

Graduate Research Assistant 2024–present
University of Colorado Boulder

Scientific Programmer / Analyst 2023–2024
NASA Goddard Institute for Space Studies

Scientific Programmer 2021–2023
NASA Goddard Institute for Space Studies

Graduate Research Assistant 2021
International Research Institute for Climate and Society

Graduate Research Assistant 2021
Sabin Center for Climate Change Law

Graduate Research Assistant 2020–2021
Lamont Doherty Earth Observatory

Peer-Reviewed Publications

Published/Accepted

12. **Elling, M.T.**, Karnauskas, K., Kowalcyk, M. et al.: Tropical oceans drive variability in soil moisture and malaria in Malawi. *Communications Medicine* (2026). https://doi.org/10.1038/s43856-026-01681-9
11. **Elling, M.T.**, Ruane, A.C., De Mel, M., et al.: An Impact-Driven Framework for Climate Model Evaluation. *Climatic Change* (2026). https://doi.org/10.1007/s10584-026-04157-w
10. Amooli, J.A., Miller, R.L., Tsigaridis, K., Chowdhury, S., Zhang, Y., Toolan, C.A., Ahmadi, S., Allen, R.J., **Elling, M.T.** et al.: Multi-Model Impacts of Dust on African Air Quality and Mortality under Regional and Global Anthropogenic Aerosol Changes. *Journal of Geophysical Research: Atmospheres* (2026). https://doi.org/10.1029/2025JD046135
9. Stjern, C.W. et al.: From muted to rapid surface warming over India under changing aerosol emissions. *CE&E*.
8. Jeon, J.-G., Allen, R.J., Yeh,S.-W., Park, C.-E., Wilcox, L.J., Samset, B.H., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.** et al.: Regional Aerosol Emission Reductions Alter the Tropical Circulation. *Environmental Research: Climate* (2026). https://doi.org/10.1088/2752-5295/ae9c63
7. Bhandekar, A., Wilcox, L.J., Lawrence, B.N., Abraham, N.L., O'Connor, F.M., Samset, B.H., Allen, R.J., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.** et al.: South Asian monsoon response to regional aerosol emission reductions: insights from RAMIP. *Environmental Research: Climate* (2026). https://doi.org/10.1088/2752-5295/ae7fad

6. Allen, R.J., Carson, T., Liu, W., Wilcox, L.J., Samset, B.H., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.** et al.: AMOC weakening in response to global and regional reductions in aerosol emissions. *Environmental Research: Climate* (2026). https://doi.org/10.1088/2752-5295/ae63ef
5. Allen, R.J., Wilcox, L.J., Samset, B.H., Ahmadi, S., Ekman, A.M.L., **Elling, M.T.** et al.: Decomposing the global and regional aerosol effective radiative forcing associated with strong versus weak air quality policies by Mid-21st Century. *Environmental Research: Climate* (2026). https://doi.org/10.1088/2752-5295/ae5418
4. Kramer, S.M., Karnauskas, K.B., **Elling, M.T.** et al.: ColdBlobMIP: A Multi-Model Assessment of the Atmospheric Response to the North Atlantic Warming Hole. *Geophysical Research Letters* (2025). https://doi.org/10.1029/2025GL117784
3. Samset, B.H., Wilcox, L.J., Allen, R.J., Stjern, C.W., Lund, M.T., Ahmadi, S., Ekman, A., **Elling, M.T.** et al.: East Asian aerosol cleanup has likely contributed to the recent acceleration in global warming. *Communications Earth & Environment* (2025). https://doi.org/10.1038/s43247-025-02527-3
2. Nazarenko, L.S., Tausnev, N.L., and **Elling, M.T.**: Temperature and Precipitation Extremes Under SSP Emission Scenarios with GISS-E2.1 Model. *Atmosphere* (2025). https://doi.org/10.3390/atmos16080920
1. Yao, Y., Ducharne, A., Cook, B.I., De Hertog, S.J., Aas, K.S., Arboleda-Obando, P.F., Buzan, J., Colin, J., Costantini, M., Decharme, B., Lawrence, D.M., Lawrence, P., Leung, L.R., Lo, M.-H., Devaraju, N., Wieder, W.R., Wu, R.-J., Zhou, T., Jägermeyr, J., McDermid, S., Pokhrel, Y., **Elling, M.** et al.: Impacts of irrigation expansion on moist-heat stress based on IRRMIP results. *Nature Communications*, (2025). https://doi.org/10.1038/s41467-025-56356-1

Submitted

Elling, M.T., Karnauskas, K.B., et al.: Future multi-hazard climate-health projections from CMIP6 using the Global Climate Health Index. *Earth's Future*.

Seng, B. et al.: East Asian Anthropogenic Aerosols Strongly Influence past and present Southern African Hydroclimate and Ecosystem Changes. *ACPD*.

Yu, X. et al.: Recent East Asian Aerosol Reductions Weaken Tropical Cyclone Intensity Over the Western North Pacific. *npj C&AS*.

Amooli, J.A. et al.: Near-term anthropogenic aerosol reductions under a future sustainability pathway could avert nearly 40% of mid-century PM2.5 mortality. *Earth's Future*.

In Preparation

Elling, M.T. and Karnauskas, K.B.: On climate-health hazard exposure in Africa and the influence of climate variability.

Elling, M.T., Muhammad, A., Karnauskas, K.B.: gchi: A Python package to calculate global climate health hazards.

Elling, M.T. Karnauskas, K.B., et al.: Historical and future projections of malaria burden in southeastern Africa based on climate-incidence relationships.

Kowalcyk, M., **Elling, M.T.**, et al.: A Multi Country Analysis of Resiliency to Climate Driven Malaria Risk.

Kowalcyk, M., **Elling, M.T.**, et al.: Comparing Machine Learning, Spatio-Temporal, and Bayesian Models for Malaria Prediction in Southeastern Africa.

Conference Papers

Cromp, S., Rabiei, B., **Elling, M.T.**, Herron, A., Hendrickson, M.: Climate PAL: Climate Analysis through Conversational AI. ml4physicalsciences.github.io/2024/files/NeurIPS_ML4PS_2024_14.pdf

Presentations (First Author & Presenter)

Interdisciplinary methods in climate and health research. CU Boulder Sustainability Research Summit, 2026. *(Oral; Invited)*

Tropical oceans drive variability in soil moisture and malaria in Malawi. European Geophysical Union, Annual Meeting 2026. *(Oral)*

Tropical oceans drive Malawi's malaria risk. American Geophysical Union, Annual Meeting 2025. *(Poster)*

Tropical oceans drive Malawi's malaria risk. C4Health Symposium, 2025. *(Oral)*

Tropical oceans drive Malawi's malaria risk. Annual Earth System and Space Science Poster Conference 2025. *(Poster)*

Enhancing Earth System Model Development Through Climate Impact-Driven Model Evaluation. American Geophysical Union, Annual Meeting 2024. *(Poster)*

A Systematic Framework for Investigating Climate Impact-Drivers in NASA GISS ESM Configurations: Towards Improving ESM Development for Impact Applications. American Geophysical Union, Annual Meeting 2023. *(Poster)*

Peer Review

Climatic Change; *Environmental Research: Climate*; *Environmental Research: Health*; *Geophysical Research Letters*; *International Journal of Biometeorology*

Teaching Experience

Guest Lecturer, Physical Oceanography & Climate — University of Colorado Boulder (Fall 2027)
Lectured on western intensification: Sverdrup & Potential Vorticity

Teaching Assistant, Polar Ice and Climate — University of Colorado Boulder (Spring 2026)
Lectured on Snowball Earth and idealized modeling approaches for evaluating Snowball Earth conditions; created coursework and led laboratory sessions; co-developed experimental materials for measuring snow albedo and physical snow properties; guided students through hands-on experiments

Lecturer, Python Bootcamp for Earth Science — University of Colorado Boulder (Aug 2026)
Co-designed and lectured a technical curriculum focused on high-dimensional data analysis and visualization using xarray and cartopy

Guest Lecturer (invited), Environmental Policy — University of Colorado Boulder (Fall 2025)
Lectured on the intersections of climate science, policy, and society, and opportunities for strengthening connections across these domains

Mentoring

Zoë Pekarek, undergraduate researcher, University of Colorado Boulder (co-mentor; 2025)

Behrad Rabiei, intern, NASA GISS (co-manager; 2023)

Sonia Cromp, intern, NASA GISS (co-manager; 2023)

Shreya Chandra, intern, NASA GISS (manager; 2022)

Service

Co-founder, Climate Action Knowledge Exchange (CAKE) (2025–present)

Member, Atmospheric and Oceanic Sciences Curriculum Committee (2025–present)

Member, Atmospheric and Oceanic Sciences Industry Night Committee (2025–present)

Reviewer, Undergraduate Research Opportunities Program (UROP) proposals (2026)

Organizer, New Era Network for Societally-Integrated Climatology (2022–2023)

Awards, Grants, Competitive Programs

Atmospheric and Oceanic Sciences Outstanding Service Award, $500 (2026)
Atmospheric and Oceanic Sciences Teaching Assistant Award (2026)
University of Colorado Boulder SOAC/ASSG, $3673 total (to CAKE, co-PI; 2025/2026)
Annual Earth System and Space Science Poster Conference: 3rd Place (2025)
ATOC Fellowship, $3627 (2025)
ATOC Fellowship, $4350 (2024)
U.S.–Korea Young Climate Activists Exchange Program (2022)
NCAA Academic All-American (2019)

Technical Reports

Carr, E., McCabe, S., **Elling, M.T.**, and Martínez-Loustalot, P.: Emissions and Underwater Noise from LNG Carrier Vessels — A Study of the Potential Impacts of LNG Development on Marine Mammals in the Gulf of California.

Carr, E., McCabe, S., and **Elling, M.T.**: Well-to-Tank Carbon Intensity of Canadian LNG.

Carr, E., McCabe, S., and **Elling, M.T.**: The Impact of Ship Emission Fees on Mode Shift Potential in the United States.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: Well-to-Tank Carbon Intensity of European LNG Imports.

Carr, E., Winebrake, J.J., McCabe, S., **Elling, M.T.**, and Green, E.H.: Ocean-Going Vessel Decarbonization Technology Assessment.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: LNG as a Marine Fuel in the United States.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: LNG and Shipping in the Arctic.

Comer, B., McCabe, S., Carr, E., **Elling, M.T.**, Sturrup, E., Knudsen, B., Beecken, J., and Winebrake, J.J.: Real-world NOx emissions from ships and implications for future regulations.

Carr, E., McCabe, S., **Elling, M.T.**, and Winebrake, J.J.: Options for Reducing Methane Emissions from New and Existing LNG-Fueled Ships.

Carr, E., Winebrake, J.J., McCabe, S., and **Elling, M.T.**: Black Carbon Emissions from Ships in the Arctic 2019–2024.

Media Coverage

Distant Oceans May Help Predict Malaria Risk in Malawi. *Eos*.

From pathogens to prehistory: A selection of recent research from Africa. *Nature Africa*.

Tropical ocean temperatures drive changes in malaria cases in Malawi. *CIRES*.

Technical Skills

UNIX/Linux and shell scripting; High-performance computing; Version control (Git); PostgreSQL and Relational Database Management

Models: NASA GISS ModelE; GEOS-Chem; InMAP

Open Source Software: GCHI (lead developer)

Programming languages: Python, Fortran, JavaScript, MATLAB, R

Professional Affiliations

American Geophysical Union (2022–present)